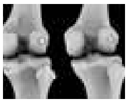

The resurgence of the Fabella
An ancient bone thought lost is making a comeback since the 1900s, and researchers are finding out why. **https://dgit.in/apr20-61**

Whales and Dolphins
Marine mammals made the trip from water to land to water, transforming their skeletons in interesting ways. **https://dgit.in/apr20-62**

theory being how embryos of different organisms develop in different ways. This mistaken understanding, at one time called Haeckel's law, had coloured the interpretations of fossil evidence by scientists for decades. It is only recently that the advancement of genetics, in general, and evolutionary genetics, in particular, that have allowed scientists insight into the matter of exactly how the skeletal system emerged. This has allowed scientists an unprecedented amount of insight into the evolution of animals. For example, scientists now know that the genes responsible for the formation of bones, they also moderate how thick the skin should be, and also regulate the growth of hair follicles. This, in turn, informs scientists that all three attributes emerged at around the same time in animals, allowing them to paint a clearer picture about the evolution of living things.

Broadly speaking, the development of the endoskeleton can be split into four stages. The very first creatures to have a structure that evolved into a skeleton were believed to be ancient forms of cephalochordata, filter feeders that fed on microbes similar to the modern day lancelets. These creatures, along with other invertebrates, have a structure called the notochord, a bunch of nerves running down the back of the organism. In vertebrates, the embryonic notochord develops into the vertebral column. Then, there was the soft cartilaginous endoskeleton that is seen in lampreys and sharks. Then came the hard exoskeleton shells of primitive fish, jawless fish with armour plating, known as Ostracoderms. Then came the fish with mineralised endoskeletons, the first ones with bones as we know them today. This was because of a process of mineralisation, where calcium compounds are deposited on the skeleton, a process that makes the bones harder. Eventually, these kinds of bony fish gave rise to the terrestrial vertebrate endoskeleton.

For millions of years, the mineral used to harden the skeleton was calcium carbonate, which was dumped into the ocean because of tectonic activity around 1.5 billion years ago. This influx of calcium carbonate allowed animals to utilise it in a number of ways, including forming the early shell like exoskeleton, as well as evolving spikes, sclerites and plates for protection. However, vertebrates replaced the calcium carbonate with calcium phosphate for the mineralisation process. It is still not known exactly why this occurred, but scientists have a few ideas. One of these is that a phosphate-based skeleton has chemical properties more suitable to maintaining their structural integrity during interactions with acids produced by the body, especially during periods of high metabolism. This theory was tested with implanting fish with carbonate and phosphate crystals in various parts of the body, and then putting the fish on various dietary regimes. In the fish with the higher metabolism, the implants had dissolved further, hinting that a phosphate-based bone mineralisation process is indeed more stable in animals with a high metabolism. In short, the move from a calcium carbonate-based mineralisation process to a calcium phosphate-based mineralisation process allowed organisms to extract more energy from their surroundings.

A fossilised trilobite, a marine arthropod (invertebrate organisms with segmented bodies, and an exoskeleton) that emerged in the Cambrian explosion

The development of mineralisation of the skeleton, as well as the exoskeleton evolving into an endoskeleton were important for a number of reasons. They allowed organisms to move over greater distances, because of the associated development of strong limbs with muscles. They also allowed the development of a sensitive skin that regenerates, instead of a hard shell that keeps building up. The animals could also find new niches for habitation, and there, evolve specialised organs and appendages. Around 0.5 billion years ago, the Earth witnessed an unprecedented evolutionary diversification event known as the Cambrian explosion. There was a sudden and dramatic change in the fossil record at this point. Animals became complex, and were seen in more different forms than ever before or since. The Cambrian explosion also saw the first calcifying animals, or creatures that built shells, There is some evidence of pre-Cambrian exoskeletons, but these thin shells are not well preserved. The emergence of the skeleton significantly helped towards the diversification of life.

One of the problems here is figuring out which came first, the hard external shells, or teeth. The question is did predators evolve teeth to counter the hardening shells of prey species, or did the prey species evolve hard shells to counter the evolution of teeth in predators. Hardening of the skin and the development of teeth are both regulated by the same set of genes. The same genes appear to be a modified form of the genes that form the template for the growth of taste buds on the tongue. This finding indicates that the machinery to develop both the shell and the tooth was already there, and only needed to be modified. Irrespective of what came first, it was the pressures of finding prey and getting predated that caused the evolution of both the hard exoskeletons and the teeth. Now, the mechanisms for developing the teeth are important because that is where the mineralisation process bagan, and the same mechanism could then be utilised elsewhere in the body, that is the development of the endoskeleton.